ARTICLE

Neuromechanics of the Rate of Force Development

Alessandro Del Vecchio
Department Artificial Intelligence in Biomedical Engineering, Friedrich-Alexander Universität, Erlangen-Nürnberg, Erlangen, Germany

DEL VECCHIO, A. Neuromechanics of the rate of force development. *Exerc. Sport Sci. Rev.*, Vol. 51, No. 1, pp. 34–42, 2023. *The rate at which an individual can develop force during rapid voluntary contractions can be influenced by both the neural drive to a muscle and its intrinsic musculotendinous properties. We hypothesize that the maximal rate of force development across human individuals is mainly attributable to the rate of motor unit recruitment.* **Key Words:** rate of force development, motor unit, motor neuron, motor control, neuromuscular physiology

Key Points

- The rate of force development of human individuals varies greatly across contractions and muscles.
- Here we hypothesize that the rate of motor unit recruitment dictates the inter-subject variability in generating force rapidly.
- Moreover, we discuss evidence that the neural drive to the muscle and the muscle components during rapid contractions behave in a parallel. The tight link between neural and muscular properties determines the association between neural drive and rate of force development during the early and late phases of force generation.
- We discuss the current evidence from computer simulations and experimental recordings of voluntary force levels and motor unit discharge characteristics during rapid contractions.

INTRODUCTION

Evolution by natural selection has developed biological systems that can produce high levels of muscle force in less than 100 ms (1–6). Despite this small time frame, there are large differences between individuals in the maximal rate of force development (1–6). By combining information from motor unit recordings, measurements of whole-muscle force, and computer simulations, we can identify some of the neuromechanical explanations for the variability in rate of force development across individuals (7–10). Here we hypothesize that the variability across human individuals is attributable mainly to differences in how fast motor units are recruited (defined thereafter as rate of motor unit recruitment) and not the rates at which they discharge action potentials (8,11–15).

Some mechanisms previously have been tested indirectly (see reference (1) for review) and suggest specific neural and muscular contributions (3,4); however, these arguments have been evaluated with methods that only capture an indirect view of the neural and musculotendinous factors that contribute to the mechanistic events associated to the fast generation of force. For example, several previous studies (see reference (1) for review), suggest specific neural versus muscular contributors during rapid force production. These studies indicate that the neural activity during the early phase of contraction (<50 ms) represents the main contributor to the rate of force development, and that during later phases (>50 ms), the muscle intrinsic properties are the predominant factor for the subsequent generation of force. The separation of neural versus muscle components is challenging for several methodological and physiological reasons. Moreover, it is likely that the neural stimulus and muscular apparatus behave in parallel.

At the physiological level, the neural and muscular apparatus in the mammalian skeletal muscle seems more likely to follow a continuum of properties and distribution, where the activity of the alpha motor neuron shapes the histological and biochemical properties of the muscle fibers (11,12,16). The parallelism between neural drive and rate of force development ((13) and later paragraphs) suggests strong matching between motoneuron properties and muscle speed characteristics (16). However, there is also evidence where the peripheral neuromuscular system can perform relatively high rates of force developments, despite a limited initial neural drive, as evidenced by high rates of force developments during supramaximal percutaneous nerve stimulations and low surface electromyography (EMG) amplitude activity (3). This suggests that there are few individuals with a low electromyography (EMG) amplitude despite a normal peripheral neuromuscular output (total muscle force). One potential explanation for this phenomenon is that the generation of rapid force contractions may have an innate or learned neural component (*e.g.*, after training) during voluntary movements.

Address for correspondence: Alessandro Del Vecchio, PhD, Department Artificial Intelligence in Biomedical Engineering, Faculty of Engineering, Friedrich-Alexander University Erlangen-Nürnberg, Zentralinstitut für Medizintechnik (ZiMT), Henkestraße 91, 91052 Erlangen, Germany (E-mail: alessandro.del.vecchio@fau.de; https://www.nsquared.tf.fau.de/).
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From a methodological perspective, it is challenging to isolate the neural and muscle component because of our inability to extract motor unit recruitment thresholds and the mechanical properties of the individual motor units. However, it is possible to indirectly infer the functional behavior of neural and muscular structures with combination of electrical percutaneous stimulation of the nerves, computational motor unit models (see paragraph on neural vs muscle contributors), and *in vivo* measures of whole muscle force and motor unit population behavior. The separation of early versus late rapid force production in time windows is also sensitive to intrasubject variability since the time windows do not represent a specific physiological mechanism but rather an arbitrary threshold. For example, some individuals with high initial rate of force development reach peak muscle forces earlier, which further suggests that rapid force contraction occurs across a wide range of speeds in different individuals, and therefore should be treated as a time series and not under sampled at a lower resolution during offline signal analysis (2,4,14,15). Currently, motor unit models can provide the only compelling evidence for understanding the neural versus muscular determinants, since they allow the tuning of neural and muscle parameters based on experiments recording whole muscle force and motor unit discharge characteristics. The combination of evidence from electrical stimulation studies, computational neuromuscular models, and the *in vivo* discharge times of motor units, indicates that the rate of motor unit recruitment is the main determinant of rate of force development across human individuals, as discussed in the later sections.

Neural Control of Rate of Force Development

The maximal rate of force development primarily depends on the initial efferent spinal drive or "neural drive" to the muscle (Fig. 1). Experimental observations of motor unit population behavior allow us to understand the time course of the recruitment and discharge rate of motor units within and between human individuals.

For example, although previous research suggested that different time points of the force-time curve are attributed to predominant neural or muscular components (see reference (1) for review), these studies associated global EMG estimates to whole muscle forces. By looking at the behavior of the motor neurons it is possible to observe that the maximal variability in rate of force production is determined mainly by the initial motor unit discharge rate and the rate of motor unit recruitment. After the initial phase (~35 ms), the motor unit discharge characteristics do not predict the inter-individual variability in rate of force production. Although intuitively this may be assumed that there are other components after the initial phase of rapid force production such as intrinsic muscular properties (1,4), we found that the variability across human individuals in motor unit discharge rate is the same after the motor units reach their plateau

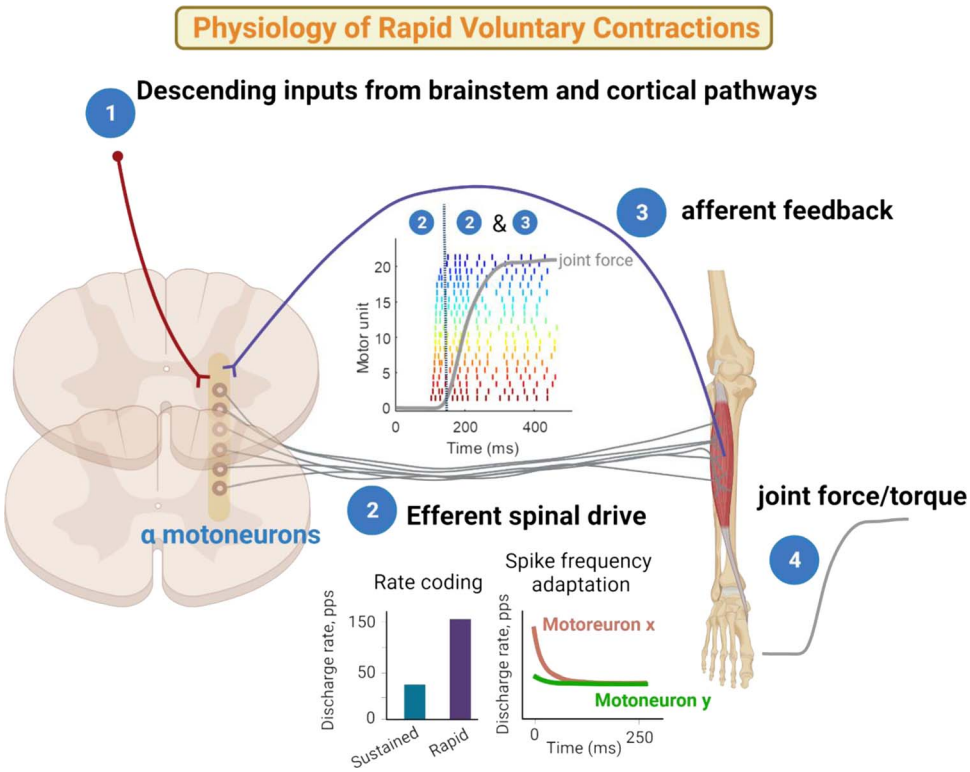

Figure 1. Descending inputs from the cortex and brainstem initiate voluntary movement [1]. The activity of the motor units in the first few milliseconds represents the influence of supraspinal inputs onto the motor neurons. Motor units are recruited at fast rate of motor unit recruitments (see the recruitment intervals before the dashed line in the raster plot 2) and the discharge rate is significantly greater during rapid contractions than during the sustained contractions (bar plot, 2). The behavior of the motor units in an individual motor nucleus seems to not behave uniformly. Some motor neurons show the classic spike-frequency-adaptation observed in *in vitro* experiments in rats and feline motor neurons. Another population of motor units does not exhibit the nonlinear decrease in discharge rate, but rather a 'flat-component' (see Fig. 5 for more details and paragraph on the spike frequency adaptation). After the first milliseconds of firing the discharge rate may decrease also because of the afferent volleys [3]. The combined activity of efferent and afferent input determines the total joint torque. The maximum derivative of joint torque (rate of force development) of a human individual can be predicted by extracting the rate of motor unit recruitment (13).

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discharge rates (Fig. 1, data set published (13)). It is indeed possible to predict the late phase force levels during maximal rate of force development contractions from the force values during the early phase of contraction (see later section and Figs. 2 and 3). For example, motor neurons showing onefold up to fivefold differences in initial discharge rates, the nonlinear decreases due to intrinsic motoneuron properties (see *Spike frequency adaptation paragraph* and Fig. 1) is followed by a plateau level that is similar across motor neurons and human individuals. Therefore, previous studies that did not find any relations between EMG amplitude and force after the initial phase of contraction (4) is likely because of the plateau levels of motor unit discharge rates (Fig. 2). Moreover, it cannot be excluded that some individuals that show low EMG amplitude values during the later phase of contraction may also have specific muscle tissue properties that contribute to a lower muscle force.

The plateau levels of motor unit discharge rates should not indicate that the neural input after the initial phase is not a determinant factor for rate of force development, but rather indicates that the high initial recruitment and discharge rate are essential for switching the muscle from a resting state to maximal generation of motor unit twitches (Fig. 2). It is clear that any relations between neural and muscular contributors must take into account the behavior of motor units. Figure 2 shows the physiological mechanisms underlying the generation of force and the parameters of an integrate-and-fire model that can be used to tune the neural (synaptic input strength and therefore neural drive to the muscle), as well as the musculotendinous output by changing the properties of the motor unit twitch parameters. Modeling the motor unit parameters with computer simulations is exceptionally relevant to understand the specific contributors of different mechanisms, such as rate of motor unit recruitment (8), discharge rate (8,10), and synchronization (22), as discussed in the next paragraph.

Motor Unit Behavior During Rapid Force Contractions: Experimental Recordings and Computer Simulations

The first seminal works that highlighted the way the nervous system controls muscle force during rapid fore contractions were accomplished in the 1970s by Desmedt and Godaux (5,23,24). Figure 3A–C shows how the central nervous system controls the recruitment and rate coding of an individual motor unit. Although the control of rapid force contraction involves several experimental paradigms, such as external sensory and auditory cues, electrical/magnetic submaximal, and maximal stimulations, the way nervous system controls muscle force can be estimated from the activity of motor units.

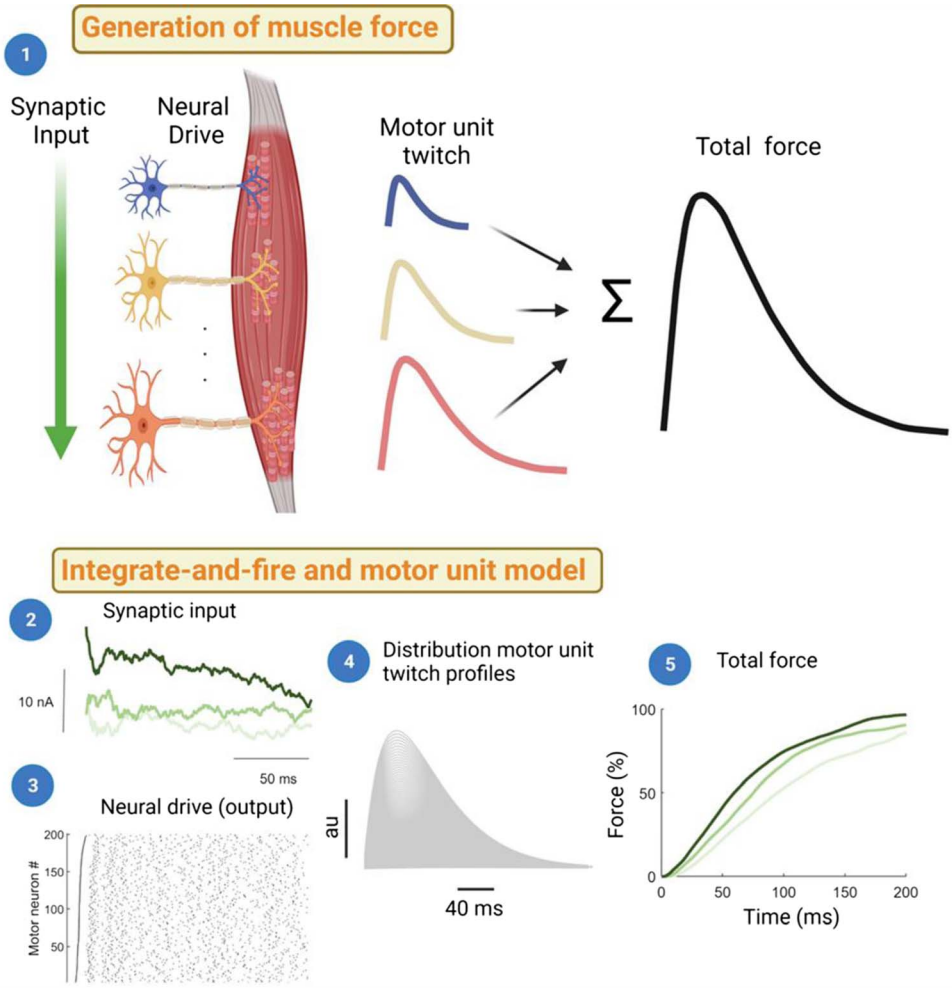

Figure 2. Generation of muscle force from synaptic input to total muscle force [1]. The neuromuscular system can be modeled with a integrate-and-fire and motor unit model [2]. Through computer simulations we can understand the relative influence of synaptic input to populations of motor neurons [2], the recruitment and discharge characteristics of the motor units [3], and the distribution of the motor unit twitches [4] on the resulting whole-muscle force [5]. In these simulations, the motor unit twitch force is estimated from experimental values reported in (19) and potentiated according to the equation reported previously (8).

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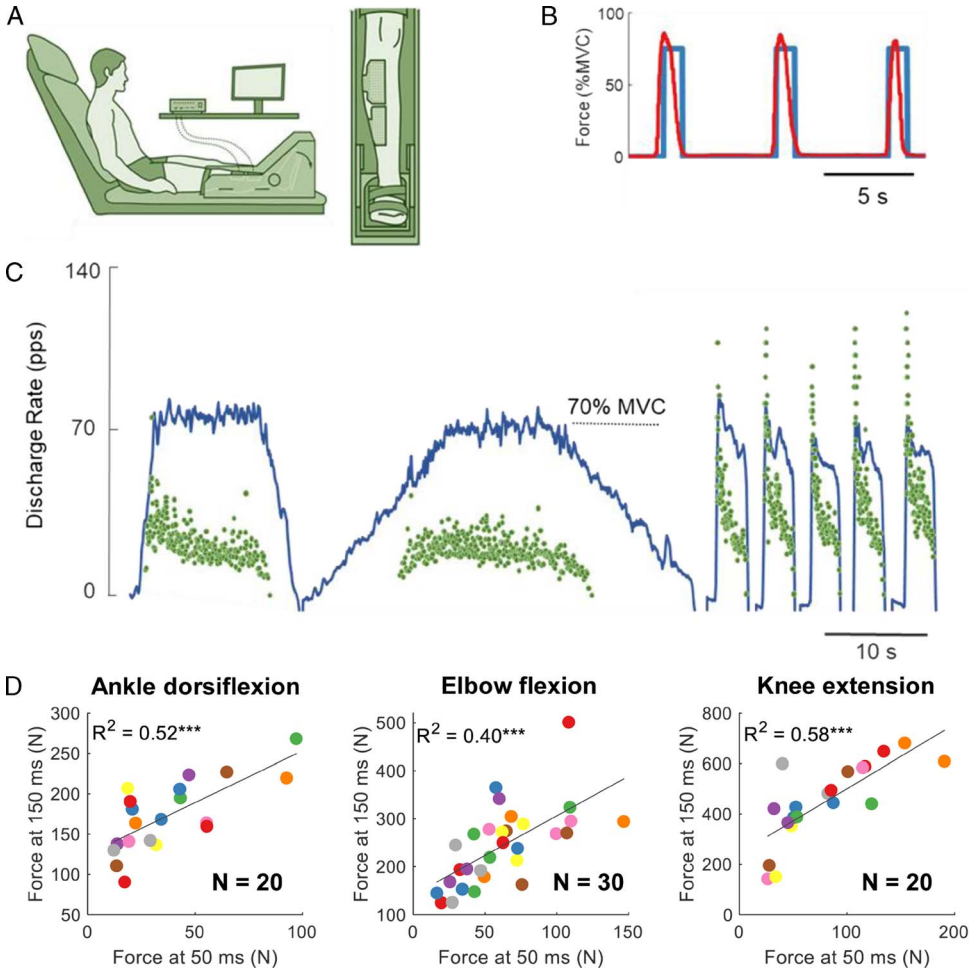

Figure 3. Neural control of the rate of force development by the nervous system. (A) A subject performing rapid and sustained isometric contractions with the ankle dorsiflexors. (B) An example of three rapid isometric contractions. The semi-rectangular line indicates the requested force trajectory and the other line indicates the acquired force data. (C) Seven isometric contractions (continuous line) and the instantaneous discharge rate of an individual motor unit that was tracked across the contractions. Note the leftward shifts in recruitment thresholds and changes in discharge rate with an increase in contraction speed, despite similar target forces. D. The association between voluntary force level at 50 ms from contraction onset and force at 150 ms for three muscle groups. Each dot represents one subject. The ankle-dorsiflexion force data (13), the elbow flexion (20), and knee extensors (21) was analyzed with the same methods and collected in two different laboratories (University of Rome "Foro Italico" and Loughborough University, the references (13,20,21) contains the full details of the anthropometrical characteristics of the subjects and their physical activity levels).

Motor unit discharge characteristics during rapid force contractions show large differences between human individuals only in the initial phase of contraction (~first 35 ms). For example, the peak discharge rate of the fastest subject out of 20 men corresponded to 200 Hz, which is 100 Hz higher than the discharge rate of slowest subject (~100 Hz). Similarly, the fastest subject showed twofold higher motor unit recruitment rate.

The nervous system controls the rate of force development by modulating the recruitment and discharge rate of motor units. Figure 3A–C shows the changes in recruitment and rate coding of an individual motor unit. In Figure 3, the discharge rate of the representative tibialis anterior motor unit was tracked across contractions at slow (5%MVC/s), relatively fast (20% MVC/s), and maximal "feedforward" contractions. The recruitment threshold of the motor unit during the slowest contraction is around 35% of maximal voluntary force, whereas during the contraction at mid-rate of force development, the recruitment threshold is around 15% of maximum. During the contractions at the maximal rate of force development (right side of the figure), the motor unit is recruited even before the generation of force, i.e., during the neuromechanical delay. As discussed in the introduction, there are several previous reports (1,3,4) that suggest that the early phase of rapid force development (<50 ms) is dominated by neural factors and only at later time points (approximately >100 ms), the muscle is the predominant contributor to the rate of force development. Figure 3D shows correlation analyses across larger number of subjects and muscle groups. These significant relations (Fig. 3D) show a strong dependence between early-phase force levels and later phases (150 ms). Therefore, if one would hypothesize that the neural and muscle are not physiologically bound (as it may assumed as they represent two different anatomical and histological systems), they likely are behaving in parallel. However, as discussed in this article, there is evidence that the neural and muscle components are tightly linked by the behavior of the motor neurons, which shapes the acute and chronic changes in muscle force output (12,13,16). As the motor unit discharge characteristics determine the maximal rate of force development (that happens in time windows between 80 to 200 ms (13)), the neural and muscle components are likely behaving as a single entity,

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anatomically and functionally, where the neural stimulus may determine the longitudinal adaptations in the mechanical characteristics of the muscle. These are likely long-term adaptations that are mediated by the properties of the motor neurons. Current (Fig. 4D and (4,13)) and previous evidence (12) collectively suggest that the properties of the motor neurons determine the contractile speed of the muscle. However, there are likely specific motor unit discharge rate patterns that mediate the fast development of force, which are discussed in the next paragraph.

Motor Unit Discharge Characteristics That Determine the Maximal Rate of Force Development

The compression of motor unit recruitment values and increase in discharge rate represent the two main mechanisms that the nervous system can modulate the rate of force production. However, the initial rate of force development of a muscle seems to be primarily mediated by how fast motor unit are recruited (Fig. 4A–G), and the discharge rate only accounts to the maximal generation of force at later stages of the contraction (i.e., after the peak of rate of force development, Fig. 4H). Figure 4F–H shows the results from a computational model that used motor unit and whole force values of 25 human individuals. The results show a strong dependence of muscle force on the rate of motor unit recruitment of motor units (Fig. 4G) and not on the initial motor unit discharge rate (Fig. 4F). The values of discharge rate at the plateau levels explained all the variance in the force values at later phases of contraction. We found a similar mechanism during sustained sinusoidal contractions at different rates of force development (25). The central nervous system modulates the delay between the neural input and force output in a dynamic way with respect to rates at which force is developed (25). As the discharge rate did not increase during these sustained contraction at different speeds, the only way the nervous system can compress the delay between neural input and force output is to recruit higher-threshold motor unit with faster contraction times (see (25) for the full details).

Although previous studies indicated that the motor unit discharge rate or occurrence of "doublets" discharges represent the

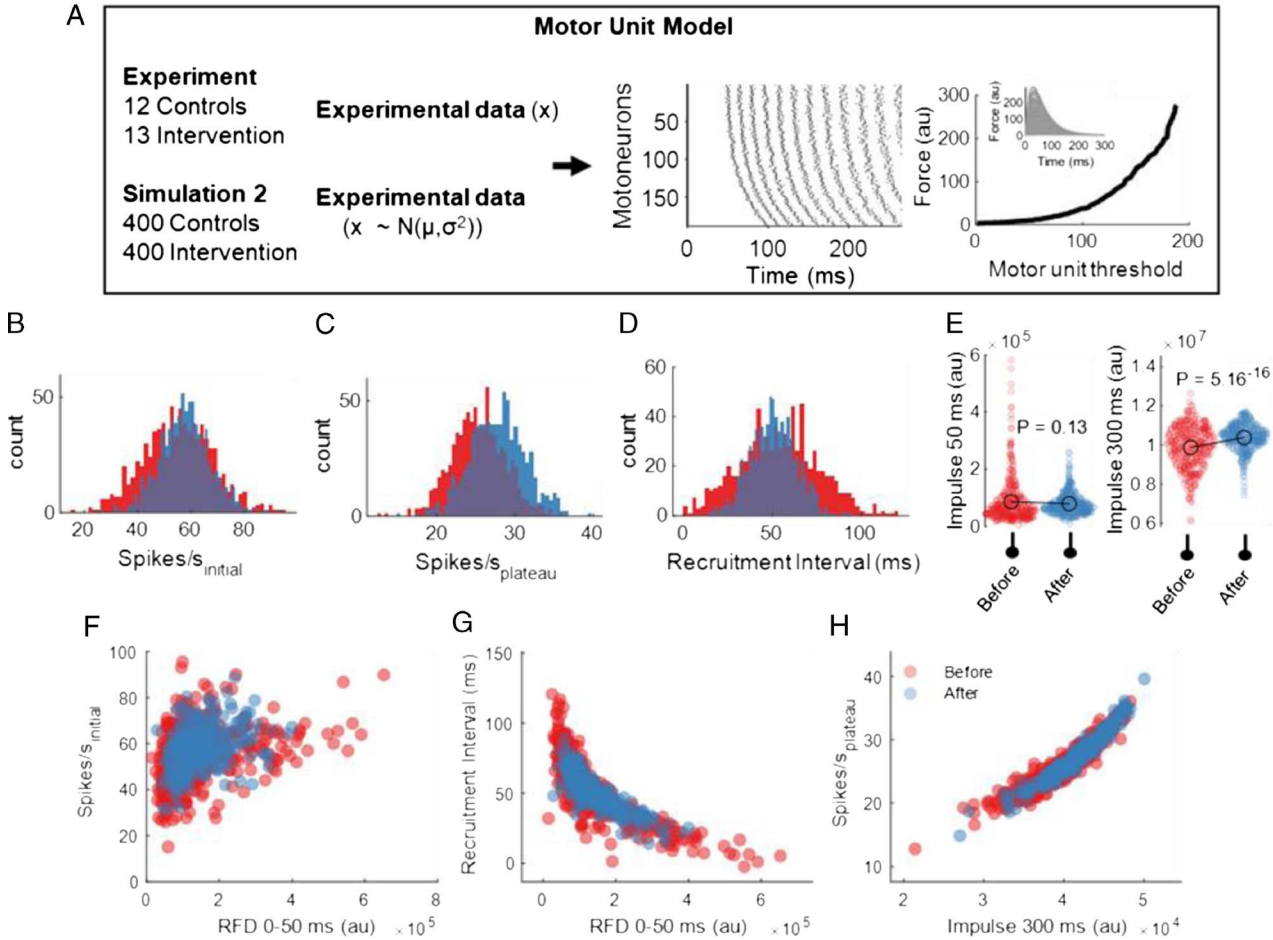

Figure 4. Training interventions that maximize strength or speed can be useful to understand the neural mechanisms underlying the maximal rate of force development. Despite significant changes in maximal force after 4 wk of training at high forces, we observed no changes in the rate of force development of the muscle. A motor unit model (A) was used to understand the motor unit adaptations that determined these changes. (B–D) Three basic motor unit parameters that have been linked to the rate of force development (initial discharge rate, plateau discharge rate, and rate of motor unit recruitment/interval, B–D, the two colors indicates before and after training). The resulting force impulse (force*time) during the early (first 50 ms) versus late (300 ms) force phase. Note that there was no change in the force impulse during the first 50 ms of contraction, because of the fact that the training intervention did not change the recruitment interval of motor neurons or initial motor unit discharge rate but only the instantaneous firing rate of motor units during the plateau phase of contraction. The correlation analysis between rate of force development (RFD) and the motor unit variables is shown in F, G, and H. Note the increase in rate of force development with faster recruitment intervals (G) but not with an increase in initial discharge rate. The strong linear relation in H shows that at high forces, the main contributor for generating high force impulse (thus high maximal voluntary force) is the maximal discharge rate of motor units. [Adapted from (10), Copyright 2022, The American Physiological Society. Used with permission.]

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predominant factor for rate of force development (26), we have now experimental and computational evidence that the rate of motor unit recruitment is the main determinant of the rate of force development. The rate of motor unit recruitment reflects the time intervals between the progressively recruited motor units. For the calculation of this value, the recruitment intervals from the first to the last recruited motor units are stored in a vector. The derivative of this vector corresponds to the time intervals separating the recruitment of consecutively detected units. For example, if in one contraction there were four identified motor units, one recruited at 2 ms, one at 10 ms, and one at 15 ms, the time intervals separating recruitment of successive units would be 8 ms (interval between the first and the second identified unit) and 5 ms (intervals between the second and the third). The average of the two intervals is 6.5 ms, and its inverse corresponds to 166.6 motor units/ms. Clearly, this measure is influenced by the number of detected motor units, which is a limitation of the approach; however, it allows to test differences within contractions (27) and also between human individuals (13). It is important to note that all current experimental values are obtained from the tibialis anterior muscle, and these values could be different across different muscles, although it is important to remember that early and late rapid force values are similar also in other muscle groups (knee extensors, and elbow flexor (Fig. 3D)). There is additional evidence from computer simulations that the rate of motor unit recruitment determines the maximal rate of force development across human individuals.

By using a similar method as described in Figure 2, we found that increasing the recruitment interval of motor units (ranging from 20 to 200 ms) shows a more than fourfold higher rate of force development than the effect of increasing the initial discharge rate (ranging from 60 to 220 pps) or doublets occurrences (8). It is possible to accurately predict the rate of force development by changing the rate of motor unit recruitment and not the discharge rate of motor units (10). Moreover, the recruitment interval and high discharge rate can compensate for slower muscle fibers contraction speeds (8). From the model, it is possible to compare the neural and muscular properties of the motor units. The modeling results show that fast recruitment of motor units and high discharge rates can compensate a slow muscle unit twitch. Specifically, increasing the rate of motor unit recruitment has a sixfold higher effect than the effect of decreasing motor unit twitch contraction times (which indirectly reflects the musculotendinous properties of the muscle).

Therefore, having a "faster" nervous system is on average six times more efficient than a fast muscle with respect to producing force rapidly. The evidence from experimental recordings and computer simulations collectively indicates that the capacity to generate rapid force production is mainly dictated by the rate of motor unit recruitment and only partly affected by muscular properties. However, as discussed in this article, these two properties follow a continuum in healthy humans, as evidenced by strong relations between maximum rate of force development and motor unit discharge characteristics (13), as well as early and late rapid force in most human muscles (Fig. 3D). Moreover, it cannot be excluded that some subjects may present a weak muscle or a strong muscle, for example after injury, disuse, and other factors. Also, there is little knowledge if potential specific muscle phenotypes shape neural structures in the spinal cord and brain.

Collectively, these results indicate that the rate of motor unit recruitment, and second, the initial motor unit discharge rate represents the main determinant of rate of force development in healthy human individuals. It is difficult to separate the motor unit discharge rate and the rate of motor unit recruitment during rapid contractions. These two mechanisms may result from a similar motor command (i.e., high excitatory and inhibitory common inputs). However, there is evidence that the spinal networks receive specific synaptic input current strengths (28), and that in humans and other nonhuman species, there are direct corticomotoneuronal connections (29) that may initiate high first spike synchrony to some motoneurons in a pool.

In the next section, we will look at what factors may determine a fast recruitment of motor neurons and high initial motor unit discharge rates.

Spinal Versus Supraspinal Activity: Cortical/Brainstem Contribution to Rate of Force Development

Mammalian motor neurons in response to supra-threshold currents show a fast decrease in the discharge rate during the first hundreds of milliseconds of firing (30). This phenomenon has been referred to as the spike frequency adaptation (SFA) and has been documented in different classes of neurons. Recent studies show that the SFA in motor neurons is associated to the availability of the fast-inactivating Na+ channels (31). Currently, there are no estimates of SFA in humans and for representative populations of motor neurons. One way to test the SFA *in vivo* relies on the identification of the discharge activity of a representative population of motor units during rapid force contractions. Indeed, rapid force contractions seem to mimic the nonlinear decrease in motor neuron discharge rate that is seen *in vitro* studies (13) (Figs. 5 and 6) probably due to suprathreshold inputs.

Figures 5 and 6 shows the time course and behavior of the motoneuron pool (242 identified neurons) in 20 human subjects. The data set of this study was published previously (13), here we report the SFA values and a blind separation approach to learn the common patterns of motor unit discharge rates during rapid force contractions. We found that the SFA ratio measured as the first two instantaneous discharge rates for a motor unit divided by the discharge rate at plateau (after the first hundred of milliseconds) explains the variability in rate of force development across human individuals. The average SFA ratio values seem to match closely the ones reported in rodents (31). Moreover, the SFA ratio was strongly correlated with the initial instantaneous discharge rate of the motor neurons (first two spikes). All human motor neurons exhibited a similar instantaneous discharge rate during the plateau phase of the firing rate and this was not predictive of the SFA.

Interestingly, the SFA varied significantly across the motor unit pool. Although we observed the classic nonlinear decrease in discharge rate as observed in *in vitro* studies in some motor units, i.e., the SFA pattern (~40% of the variance explained for all subjects), there was a strong, second "flat" component within the population of motor neurons in all human individuals (Fig. 6). This component explained the remaining variance in the behavior of all recorded neurons during the rapid isometric contractions. It seems that the human motoneuron pool in response to high synaptic inputs shows two clusters: the main

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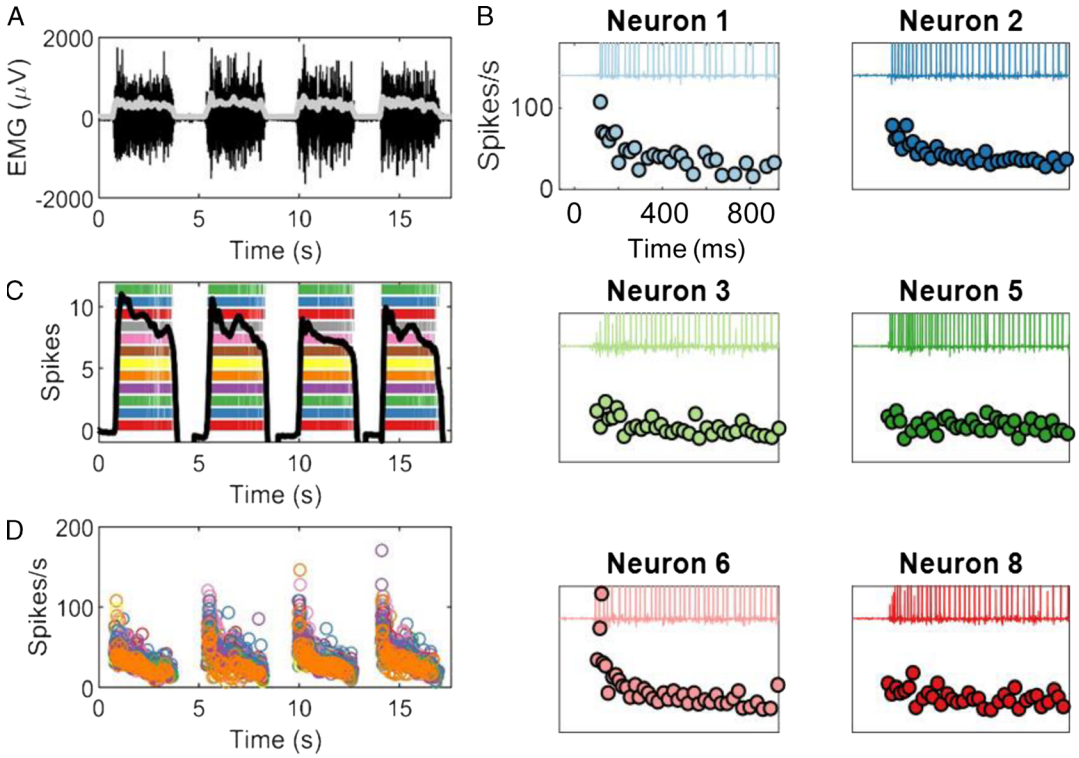

Figure 5. Spike frequency adaptation in human motor units. A. Four tibialis anterior isometric contractions (gray line represents the smoothed amplitude of the electromyography (EMG)), and one monopolar EMG signal (blacklines). (B) The EMG signals were decomposed in individual motor unit spike trains (force trace in black). (C) The instantaneous discharge rate of the motor units shown in B. (D) The instantaneous discharge rate for six selected motor units. The top traces show the separation vector, which represents an estimate of the decomposition performance. Note the variability in the discharge characteristics across motor neurons in the same motor nucleus. Some neurons show a high initial discharge rate, which mirrors closely previous finding in *in vitro* isolated motor neurons responding to suprathreshold currents [see (17) and (18)]. Other neurons, such as motor unit 3 and motor unit 5, show a constant discharge rate throughout the contraction.

component as classically reported in humans as the SFA, and another component that shows a relatively constant discharge rate. It is important to note that these two clusters seem to be fixed across the same experimental session, so that motor units that show a specific component, this is constant throughout the other rapid isometric contractions. The underlying factors that determine this divergent behavior are unknown, but this could be attributed to distinct neural modules transmitted to motor neurons by brainstem or cortical pathways. There is evidence that an auditory startling cue might increase the discharge rate of a subset motor neurons that show a relatively low initial discharge rate during rapid contractions (21), suggesting that different supraspinal inputs may mediate the neural modules observed in these two clusters. This indicates that the input may be a suprathreshold for some, but not all, motor neurons. However, the possibility that the observed output may also account at least in part to different intrinsic properties of the motor neurons or interneurons circuitries should not be excluded. Future studies should test this hypothesis directly by accounting, for example, the recruitment threshold of the motor units.

It has been recently reported that the there is a strong contribution of the reticulospinal formation to the response to loud auditory cues, suggesting that the brainstem may account significantly for the high levels synaptic input that project to spinal motor neurons (17,32). This is important because the reticulospinal axons have large diameters and thus fast propagating action potentials (33). The SFA might be an important mechanism for generating force rapidly that follows the rate of motor unit recruitment. As the rate of force development in the first milliseconds is determined by the rate of motor unit recruitment, a high first- and second-spike synchrony mediates a faster rise in force. Indeed, when the human skeletal muscle is stimulated with electrical stimulation in the frequency range matching the initial discharge rates of the motor units (200 Hz bandwidth (13)), this results in a faster rate of force development than stimulation with lower frequencies (34,35).

Collectively, these results indicate that evolution optimized the nervous system to match the properties of the musculotendinous unit. Moreover, the results on auditory stimuli also indicate that neurons in the brainstem may augment the motoneuron discharge characteristics and rate of force development. Another important mechanism that may compress the recruitment threshold of the motor units and increase the discharge rate is the amount of neuromodulation provided by monoamines (serotonin and norepinephrine) and the generation of intrinsic persistent inward currents. The effects of these neuromodulators on motoneuron behavior (input resistance, spike threshold, discharge frequency acceleration) can be augmented by serotonin and norepinephrine. A recent study suggests that the blockage of serotonin receptor suppresses the rate of force development and motor unit discharge rate (18). However, the time course of neuromodulatory inputs to motoneurons during brief and rapid movements less than 200 ms is not well understood. Also, the release and

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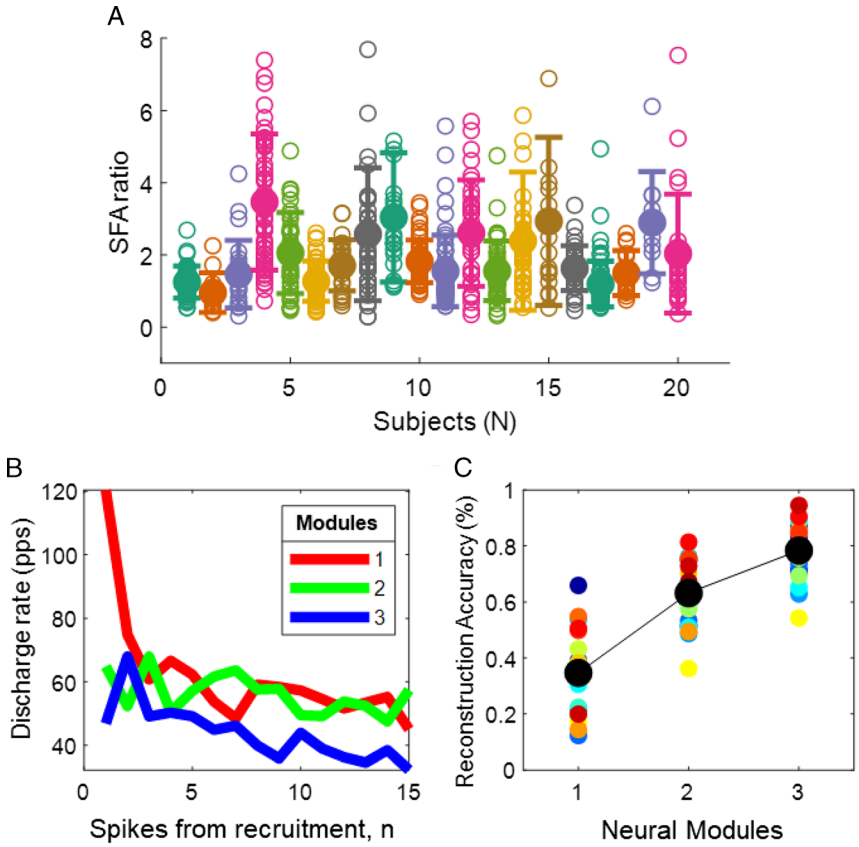

Figure 6. The variability in spike frequency adaptation (SFA), measured as the ratio between initial discharge rate and discharge rate at plateau across 20 subjects (color-coded) and motor units. (B) A factorization method (nonnegative matrix factorization) was used to identify the neural modules that explain the variance of the original signal. Note that there are two main patterns, the classic SFA pattern (high discharge rate at the start) and a flat component (the other two lines). (C) The reconstruction accuracy of these three neural modules across all subjects (each dot represents one subject).

concentration of these amines across individuals and how they relate to rate of force development is still not well known.

CONCLUSIONS

Current results provide compelling evidence that the neural drive to the muscle plays a dominant role in determining the maximum rate of force development across human individuals. The rapid force development produced by skeletal muscle can be predicted by looking at the initial motor unit discharge characteristics. It was previously assumed that the discharge rate of motor units represents the main determinant of rate of force development. Here we provide evidence that the rate of motor unit recruitment represents the primary mechanism that mediates the maximal rate of force development across human individuals. Moreover, the current results strongly support the notion that the initial neural drive to the muscle is correlated to both early and late phases of the rapid contractions. This suggests that in healthy human individuals the neural drive and muscle components follow a continuum of properties. Moreover, from a methodological perspective these results indicate that the analysis of rapid isometric force contractions should not be assessed in time windows but rather as a continuous signal. Computer simulations indicate that a "faster" nervous system is, on average, six times more efficient than a fast muscle.

Future research should look at the plasticity of the motor neurons with training, from birth to adolescence. This might be relevant since having a fast muscle is likely mediated by specific motoneuron properties. It is currently unknown if high levels motor neuron output during rapid movements can be acquired with training. For example, it could be hypothesized that the training status during development may change the intrinsic properties of the motor neurons throughout the lifespan. This could have important implications for the development of intrinsic motor neuron properties in children and adolescence but also during aging and disease.

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